Converting Between Rectangular and Polar Coordinates Answer Key

Precalculus

Quick Answers

1. (a) x = 3, (b) y = $3\sqrt{3}$
2. (a) r = $3\sqrt{2}$, (b) theta = 45 degrees
3. (a) x = $-4\sqrt{3}$, (b) y = −4
4. (a) r = 13, (b) theta = 67.38 degrees
5. (a) r = 4, (b) theta = 120 degrees, —
6. (a) x = $-2\sqrt{2}$, (b) y = $-2\sqrt{2}$
7. (a) the angle in the negative-r name = $\frac{5\pi}{3}$, (b) x from the negative-r name = $-\frac{5}{2}$
8. (a) r = $6\sqrt{2}$, (b) theta = 225 degrees
9. (a) r = 7, (b) check that theta = 270 degrees gives back y = −7, —
10. (a) distance r, in km = 15 km, (b) bearing angle theta = 126.87 degrees, (c) angle in the negative-r name = 306.87 degrees, —

What is verified

21 of 24 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 5, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus
HSF-IF.C.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 24 tagged checks.

1. Polar $(6, 60°)$ to rectangular.

(a) $x = r\cos\theta = 6\cos 60° = 6\left(\frac{1}{2}\right)$. $\boxed{x = 3}$

(b) $y = r\sin\theta = 6\sin 60° = 6\left(\frac{\sqrt{3}}{2}\right)$. $\boxed{y = 3\sqrt{3}}$

2. Rectangular $(3, 3)$ to polar.

(a) $r = \sqrt{3^2 + 3^2} = \sqrt{18}$, and $\sqrt{18} = \sqrt{9 \cdot 2}$. $\boxed{r = 3\sqrt{2}}$

(b) $\tan\theta = \dfrac{3}{3} = 1$, and the point is in quadrant I (both coordinates positive), so the calculator's $\tan^{-1}(1)$ needs no correction. $\boxed{\theta = 45°}$

3. Polar $(8, 210°)$ to rectangular.

(a) $210°$ is in quadrant III, where cosine is negative: $x = 8\cos 210° = 8\left(-\frac{\sqrt{3}}{2}\right)$.

$$\boxed{x = -4\sqrt{3}}$$

(b) $y = 8\sin 210° = 8\left(-\frac{1}{2}\right)$. Both coordinates are negative, which is exactly what quadrant III requires.

$$\boxed{y = -4}$$

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4. Rectangular $(5, 12)$ to polar.

(a) $r = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169}$.

$$\boxed{r = 13}$$

(b) The point is in quadrant I, so no correction is needed: $\theta = \tan^{-1}\left(\dfrac{12}{5}\right) = 67.3801\ldots°$.

$$\boxed{\theta \approx 67.38°}$$

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5. Rectangular $(-2, 2\sqrt{3})$ to polar. (a) r. (b) θ. (c) Why the calculator is wrong.

(a) $r = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = \sqrt{16}$.

$$\boxed{r = 4}$$

(b) $\tan\theta = \dfrac{2\sqrt{3}}{-2} = -\sqrt{3}$, and the calculator returns $-60°$. The point has $x < 0$ and $y > 0$, so it lies in quadrant II; adding $180°$ moves the angle there: $\theta = -60° + 180° = 120°$.

$$\boxed{\theta = 120°}$$

(c) *Instructor-judged.* A correct response says $\tan^{-1}$ only ever returns angles between $-90°$ and $90°$, so its output always lands in quadrant I or IV, while this point is in quadrant II. The repair — add $180°$ whenever $x < 0$ — is justified by the period of tangent being $180°$, so two exactly opposite directions share the same ratio y/x. Reporting the corrected angle without that reason is partial credit.

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6. Polar $(-4, 45°)$ to rectangular.

(a) The formulas are unchanged by the sign of r: $x = -4\cos 45° = -4\left(\frac{\sqrt{2}}{2}\right)$.

$$\boxed{x = -2\sqrt{2}}$$

(b) $y = -4\sin 45° = -4\left(\frac{\sqrt{2}}{2}\right)$.

$$\boxed{y = -2\sqrt{2}}$$

A negative r means: face $45°$, then walk *backwards* 4 units — which lands the point in quadrant III, where both coordinates are indeed negative.

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7. Polar $\left(5, \frac{2\pi}{3}\right)$ renamed as $(-5, \theta')$.

(a) Flipping the sign of r reverses the direction, so the angle turns by a straight angle: $\theta' = \dfrac{2\pi}{3} + \pi = \dfrac{2\pi}{3} + \dfrac{3\pi}{3}$.

$$\boxed{\theta' = \frac{5\pi}{3}}$$

(b) Check with the x formula: $x = -5\cos\dfrac{5\pi}{3} = -5\left(\frac{1}{2}\right)$, and the original name gives $5\cos\dfrac{2\pi}{3} = 5\left(-\frac{1}{2}\right)$ — the same value.

$$\boxed{x = -\frac{5}{2}}$$

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8. Rectangular $(-6, -6)$ to polar.

(a) $r = \sqrt{36 + 36} = \sqrt{72} = \sqrt{36 \cdot 2}$. $\boxed{r = 6\sqrt{2}}$

(b) $\tan\theta = \dfrac{-6}{-6} = 1$, and $\tan^{-1}(1) = 45°$ — but both coordinates are negative, so the point is in quadrant III, not quadrant I. Since $x < 0$, add $180°$: $\theta = 45° + 180° = 225°$. $\boxed{\theta = 225°}$

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9. Rectangular $(0, -7)$ to polar. (a) r. (b) Check $\theta = 270°$. (c) Why $\tan^{-1}$ fails.

(a) $r = \sqrt{0^2 + (-7)^2} = \sqrt{49}$. $\boxed{r = 7}$

(b) $7\sin 270° = 7(-1) = -7$, which is the given y; and $7\cos 270° = 0$, the given x. So $270°$ is confirmed. $\boxed{-7}$

(c) *Instructor-judged.* A correct response says the formula requires the ratio y/x, and $x = 0$ here, so the ratio is undefined and the formula cannot be applied at all. The angle is read from the picture instead: the point lies on the negative half of the vertical axis, a straight angle plus a right angle round from the positive horizontal axis, so $\theta = 270°$. Saying only that the calculator errors is not full credit.

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10. Radar: aircraft at $(-9, 12)$ km. (a) r. (b) θ. (c) θ'' for the $(-15, \theta'')$ name. (d) Explain.

(a) $r = \sqrt{(-9)^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225}$. $\boxed{r = 15 \text{ km}}$

(b) $\tan^{-1}\left(\dfrac{12}{-9}\right) = -53.1301\ldots°$, and $x < 0$ puts the aircraft in quadrant II, so add $180°$: $\theta = 180° - 53.1301\ldots° = 126.8699\ldots°$. $\boxed{\theta \approx 126.87°}$

(c) Reversing the sign of the distance turns the direction through a straight angle: $\theta'' = 126.8699\ldots° + 180° = 306.8699\ldots°$, which lies in $[0°, 360°)$. $\boxed{\theta'' \approx 306.87°}$

(d) *Instructor-judged.* A correct response explains that a negative first coordinate means facing the stated direction and stepping *backwards*, which puts you in exactly the same place as facing the opposite direction and stepping forwards. Adding $180°$ to the angle and flipping the sign of the distance therefore leaves the point unmoved, so both entries log the same aircraft. Noticing only that the two angles differ by $180°$, without the reversal argument, is partial credit.